

A Minimal Coherence-to-Gain Criterion for Four-Packet Reassembly

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Abstract

TPC-278 showed that a favorable signed packet cross term can change sign under a nearby finite shell or clock choice. This paper isolates the exact finite-dimensional input that a growing source theorem would have to provide. For four vectors, let D be the packet-diagonal energy, G the energy of their sum, and E the net real cross term. We prove the exact deficit identity $\Delta = 1 - G/D = -2E/D$ and show that a power gain is equivalent to an upper bound on G/D , or equivalently to Δ being within the required power scale of one. We also prove the sharp pairwise-coherence envelope $G/D \leq \min(4, 1 + 3\mu)$, together with its reciprocal gain floor. Orthogonal packets attain the no-gain endpoint, while explicit near-canceling packets attain arbitrarily large gain. An exact interval transfer of the twelve TPC-278 source rows validates the coordinate change but supplies no asymptotic or arithmetic credit. The result is a reusable structural theorem and a precise firewall: pairwise absolute coherence alone cannot pay a positive power; a schedule-specific aggregate deficit estimate is the minimal missing hypothesis.

1 Motivation and notation

The four-packet source interface in TPC-277 and TPC-278 produces vectors V_0, V_1, V_2, V_3 in a real or complex Hilbert space. The quantities relevant to signed reassembly are

$$D = \sum_{j=0}^3 \|V_j\|^2, \quad G = \left\| \sum_{j=0}^3 V_j \right\|^2, \quad E = \sum_{0 \leq j < k \leq 3} \operatorname{Re} \langle V_j, V_k \rangle.$$

TPC-277 proved the universal floor $G \leq 4D$ and found a favorable sign on an eight-row finite source scan. TPC-278 then varied only a declared shell endpoint or clock and found four finite sign flips. Thus the next useful question is not whether a sign happens on one table, but what exact source inequality is equivalent to the desired gain.

Throughout, assume $D > 0$. When $G > 0$ write

$$r = \frac{D}{G}, \quad q = \frac{G}{D}, \quad \Delta = 1 - q = \frac{D - G}{D}.$$

2 The exact deficit criterion

Theorem 1 (minimal four-packet criterion). *For any four Hilbert-space vectors with $D > 0$,*

$$0 \leq q \leq 4, \quad G - D = 2E, \quad \Delta = -\frac{2E}{D}.$$

If $G > 0$, then

$$r = \frac{1}{q} = \frac{1}{1 - \Delta}.$$

Consequently, for $b > 0$, $\gamma \geq 0$, and $X \geq 1$,

$$r \geq bX^\gamma \iff q \leq b^{-1}X^{-\gamma} \iff \Delta \geq 1 - b^{-1}X^{-\gamma}.$$

Proof. Expanding the squared norm gives

$$G = D + 2 \sum_{j < k} \operatorname{Re} \langle V_j, V_k \rangle = D + 2E.$$

The lower bound $G \geq 0$ is immediate, while Cauchy–Schwarz gives

$$G \leq \left(\sum_j \|V_j\| \right)^2 \leq 4 \sum_j \|V_j\|^2 = 4D.$$

Division by D yields the first two identities and $q \in [0, 4]$. For $G > 0$, reciprocal division gives $r = 1/q = (1 - \Delta)^{-1}$, and taking positive reciprocals proves the three equivalent inequalities. $\square$

The theorem is an equivalence, rather than merely a sufficient estimate. In particular, the sign condition $E \leq 0$ gives only $q \leq 1$ and hence $r \geq 1$. It says nothing about how close q is to zero. A power gain requires the stronger near-cancellation scale $q \ll X^{-\gamma}$, or the equivalent deficit condition in the theorem. If $G = 0$, the sum packet is exactly zero and the same conclusion holds in the extended convention $r = +\infty$; the displayed reciprocal formula is used only for $G > 0$.

Corollary 2 (sign is not a power estimate). *No assertion of the form $E \leq 0$ alone can imply $r \geq bX^\gamma$ for any fixed $b > 0$ and $\gamma > 0$ on an unbounded scale.*

Proof. Take four mutually orthogonal unit vectors. Then $D = G = 4$, $E = 0$, and $r = 1$ at every scale. This also gives $\Delta = 0$ in the exact criterion. $\square$

3 What pairwise coherence can and cannot prove

For nonzero packet norms define

$$\mu = \max_{j < k} \frac{|\langle V_j, V_k \rangle|}{\|V_j\| \|V_k\|} \in [0, 1],$$

ignoring pairs with a zero norm (their contribution is zero). This is an absolute, phase-blind statistic.

Theorem 3 (sharp pairwise-coherence envelope). *For four packets with $D > 0$,*

$$G \leq D + 2\mu \sum_{j < k} \|V_j\| \|V_k\| \leq (1 + 3\mu)D,$$

and therefore

$$\frac{G}{D} \leq \min\{4, 1 + 3\mu\}.$$

If $G > 0$, then

$$r \geq \max \left\{ \frac{1}{4}, \frac{1}{1 + 3\mu} \right\}.$$

The constants are sharp for every $\mu \in [0, 1]$.

Proof. Put $a_j = \|V_j\|$. The definition of μ and the triangle inequality give

$$G = D + 2E \leq D + 2\mu \sum_{j < k} a_j a_k = D + \mu \left(\left(\sum_j a_j \right)^2 - D \right).$$

By Cauchy–Schwarz, $(\sum_j a_j)^2 \leq 4D$, proving the second inequality. Combining it with the universal estimate $G \leq 4D$ gives the minimum. Taking reciprocals gives the gain floor when $G > 0$.

For sharpness, let

$$\Gamma_\mu = (1 - \mu)I_4 + \mu \mathbf{1}\mathbf{1}^*.$$

Its eigenvalues are $1 - \mu$ (multiplicity three) and $1 + 3\mu$, so it is a Gram matrix for every $\mu \in [0, 1]$. Four unit vectors realizing this Gram matrix have $D = 4$, $G = 4(1 + 3\mu)$, and pairwise coherence exactly μ . Thus equality holds in the coherence envelope. At $\mu = 1$ it also attains the universal floor $r = 1/4$. $\square$

The envelope is useful as a ceiling, but it is not a signed source theorem. At $\mu = 0$ it gives $r \geq 1$, not a positive power. More generally, any uniform bound on μ leaves $1/(1 + 3\mu)$ at constant scale. The missing quantity is the signed aggregate deficit Δ , not an absolute pairwise coherence bound.

4 Sharp adversaries and near cancellation

The equicorrelated Gram family contains two instructive endpoints. At $\mu = 0$ the packets are orthogonal and $r = 1$, so no positive power can be inferred from coherence information. At $\mu = 1$ they are aligned and $r = 1/4$, showing that the four-packet universal floor is exact.

The opposite phenomenon, arbitrarily large gain, is also explicit. For $0 < \varepsilon < 1$, take scalar packets

$$V_0 = V_1 = V_2 = 1, \quad V_3 = -(3 - \varepsilon).$$

Then

$$D = 3 + (3 - \varepsilon)^2, \quad G = \varepsilon^2, \quad qquadr = \frac{3 + (3 - \varepsilon)^2}{\varepsilon^2}, \quad \Delta = 1 - \frac{\varepsilon^2}{3 + (3 - \varepsilon)^2}.$$

As $\varepsilon \downarrow 0$, the gain diverges, but $\mu = 1$ throughout. Thus large gain is compatible with maximal absolute coherence and is controlled only by the aggregate cancellation scale. The certificate uses $\varepsilon = 1/10$, for which $D = 1141/100$, $G = 1/100$, and $r = 1141$.

5 Transfer of the TPC-278 source rows

The TPC-278 certificate contains outward intervals for $r = D/G$ and for the independently replayed cancellation fraction. We transform each positive gain interval $[r_-, r_+]$ into the exact reciprocal interval $[1/r_+, 1/r_-]$ for q . The resulting deficit interval is intersected with the independently stored cancellation interval. This is a certified coordinate transfer; displayed decimals below are only midpoints.

The transfer has eight positive-deficit and four negative-deficit rows, matching the parent cross-sign census. It does not create a growing quantifier: the finite values merely illustrate the exact coordinates and the instability already certified by TPC-278. In particular, the natural row at $N = 192$ has a small positive deficit, while the neighboring shell and clock choices cross through zero.

6 Route consequence and claim firewall

The exact criterion identifies the next source-level theorem without leaving room for a hidden normalization step. To obtain a gain $r \geq bX^\gamma$, one must prove

$$\frac{G}{D} \leq b^{-1}X^{-\gamma} \quad \text{or equivalently} \quad \Delta \geq 1 - b^{-1}X^{-\gamma}$$

N	H	Q	role	$r = D/G$	$q = G/D$	Δ	sign
128	24	4	shell−	1.418569	0.704936	0.295064	+
128	24	5	shell ref.	1.873826	0.533667	0.466333	+
128	24	6	shell+	0.952448	1.049926	-0.049926	−
192	32	5	shell−	1.639087	0.610096	0.389904	+
192	32	6	natural	1.006248	0.993791	0.006209	+
192	32	7	shell+	0.866928	1.153499	-0.153499	−
256	38	5	shell−	0.912829	1.095496	-0.095496	−
256	38	6	natural	1.058744	0.944516	0.055484	+
256	38	7	shell+	1.360350	0.735105	0.264895	+
192	29	6	clock−	0.977376	1.023148	-0.023148	−
192	35	6	clock+	1.031302	0.969648	0.030352	+
384	50	7	natural	1.124747	0.889089	0.110911	+

Table 1: Coordinate transfer from the exact TPC-278 rows. The sign is the sign of the certified deficit interval, and is equivalent to the parent cross-term sign.

on the precisely declared shell, clock, masks, and packet schedule. A negative cross-term sign is strictly weaker, and pairwise absolute coherence is phase-blind. The source-level deficit estimate is therefore the minimal missing hypothesis for the signed-gain branch.

This paper proves a finite-dimensional theorem and certifies a coordinate transfer only. It does not estimate the arithmetic coefficient, prove an L^2 bound, pay the strict $1/400$ endpoint, or imply a twin-prime result. The Session-named Route-A/Route-B evaluator files are absent in this checkout; the local proof package, exact certificate, independent checker, stress audit, and Bridge-B checker supply the fail-closed scoped evaluation.

ROUTE-A	not applicable to this structural Hilbert-space theorem
ROUTE-B	yes, scoped coherence-to-gain criterion
PAIRWISE-COHERENCE-POWER	refuted exactly by orthogonal packets
FIXED-POWER-CREDIT	0
ARITHMETIC-L2 / FULL-GATE-B	open
TWIN-PRIME-RESULT	none

References

- [1] L. Wang, “Cross-Scale Signed-Gain Stability and a Shell/Clock Counterexample,” TPC-278 project release, 2026.
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- [3] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.